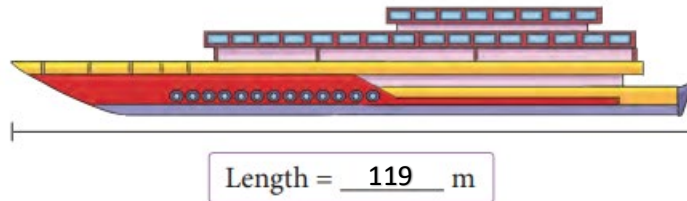
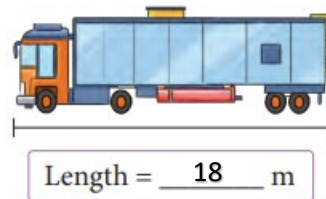
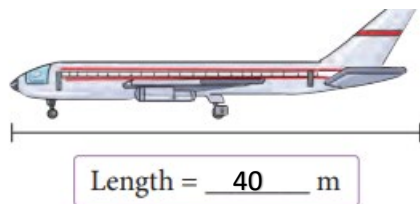


14 Perimeter and Area

Tune In



The total length of all the above transports is 177 m .

Let's Try 1

To be done by students

Let's Try 2

1. Perimeter = sum of all sides
= 4 cm + 4 cm + 7cm + 7cm
= 22 cm (Ans)

2. Perimeter = sum of all sides
= 4 cm + 4cm + 4cm + 4 cm
= 16 cm (Ans)

Exercise 14A

4. Perimeter = sum of sides
= 4 cm + 2 cm + 1 cm + 5 cm
= 12 cm

Ans: Perimeter of the figure is 12 cm

Application Based Questions

5. Length (l) of rectangle = 10 cm

Breadth (b) of rectangle = 7 cm

Perimeter = sum of all 4 sides of rectangle

$$= l + b + l + b = 10 \text{ cm} + 7 \text{ cm} + 10 \text{ cm} + 7 \text{ cm} = 34 \text{ cm}$$

Ans: The perimeter of the postcard is 34 cm.

6. Length of rectangle = 52 cm

Breadth of rect. = 52 cm – 20 cm = 32 cm

$$\begin{aligned} \text{Perimeter of rect.} &= l + b + l + b = 52 \text{ cm} + 32 \text{ cm} + 52 \text{ cm} + 32 \text{ cm} \\ &= 168 \text{ cm} \end{aligned}$$

Ans: Perimeter of the rectangle is 168 cm.

7. Distance Ritu ran around the field = Perimeter of the rectangular field
 = sum of the 4 sides of the field
 = 120 m + 80 m + 120 m + 80 m
 = 400 m

Ans: Ritu ran a distance of 400 m around the field.

8. Length of the wire required for fencing = Perimeter of the square garden
 = sum of the four sides of the garden
 = 21 m + 21 m + 21 m + 21 m
 = 4 x 21 m
 = 84 m

Ans: Length of the wire required for fencing the garden is 84 m

9. Distance Vivek cycled around the garden = perimeter of the pond x number of rounds
 = 1 km 500 m x 2
 = 1500 m x 2
 = 3000 m or 3 km

Ans: Vivek rode the bicycle for 3 km.

10. Length of the ribbon required = Perimeter of the given figure
 = sum of all sides of the figure
 = 5 cm + 4 cm + 5 cm + 1cm + 4 cm + 4cm + 4 cm + 1 cm
 = 28 cm

Ans: The length of the ribbon required to paste the ribbon is 28 cm.

11. Length of the wire required to fence the plot + shed = Perimeter of the plot + shed
 Perimeter of the plot + shed = sum of all sides forming the boundary
 = 3 x 30 m + (30 – 16) m + 16 m + 2 x 22 m (refer figure)
 = 90 m + 14 m + 16 m + 44 m
 = 164 m

Ans: The length of the barbed wire required for fencing the plot and shed is 164 m.

Exercise 14B

3.a. 8 sq. units b. 16 sq units c. 11 sq units d. 10 sq units

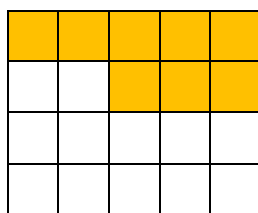
4.a. 20 sq units
 = 20 sq units
 = 20 sq. units (**Ans**)

b. 18 sq units + 4 x $\frac{3}{4}$ sq. units + 4 x $\frac{1}{4}$
 = 18 sq units + 3 sq units + 1 sq unit
 = 22 sq units (**Ans**)

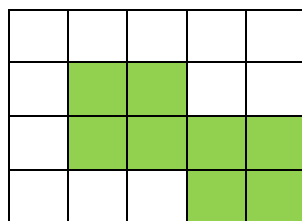
c. 20 sq units + 6 sq units
 26 sq units (**Ans**)

Application Based Questions

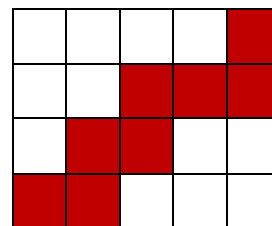
5.b.



c.



d.



Revision Exercises

Start Up

1.a. Perimeter = 18 units

Area = 15 sq. units

b. Perimeter = 18 units

Area = 12 sq units

c. Perimeter = 20 units

Area = 19 sq units

2.

No	Shape	Dimensions	Perimeter
a.	Rectangle	Length = 25 cm, breadth = 17 cm	84 cm
b.	Square	Side = 41 cm	164 cm
c.	Rectangle	Length = 15 cm, breadth = 11 cm	52 m
d.	Square	Side = 12 cm	48 m

3.a. Area = 18 sq units

b. Area = 14 sq units + $\frac{1}{2}$ sq unit = $14\frac{1}{2}$ sq units

c. 4 sq units + 8 sq units = 12 sq units.

Level Up

1.

a. Perimeter = sum of all sides = 2 cm + 4 cm + 2 cm + 4 cm + 2 cm + 8 cm + 6 cm + 8 cm
= 36 cm (Ans)

b. Perimeter = sum of all sides = 8 cm + 8 cm + 6 cm + 6 cm + 2 cm + 14 cm
= 44 cm (Ans)

2. a. Perimeter of a rectangle = Sum of all its 4 sides = $2 \times L + 2 \times b$
= $2 \times 27 \text{ cm} + 2 \times 18 \text{ cm} = 54 \text{ cm} + 36 \text{ cm} = 90 \text{ cm}$

Ans: Perimeter of the rectangle = 90 cm.

b. Length of barbed wire required = sum of all four sides – width of the gate
= $4 \times 55 \text{ m} - 6 \text{ m} = 220 - 6 \text{ m} = 214 \text{ m}$

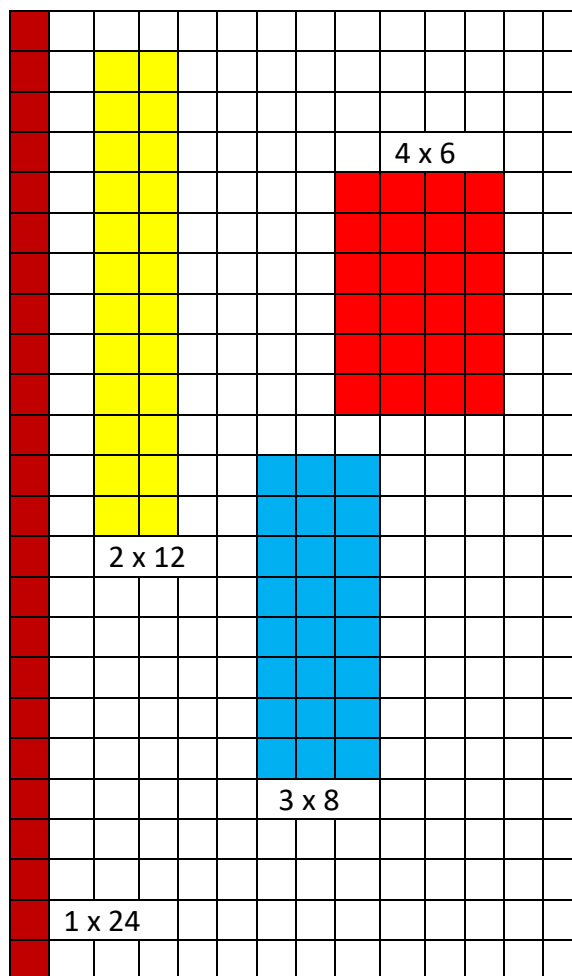
Ans: The length of the barbed wire required to fence the plot is 214 m.

c. Perimeter of a square tile = $4 \times \text{side of the tile} = 120 \text{ cm} = 4 \times \text{side of the tile}$
 $120 \text{ cm} \div 4 = \text{Side of the tile}$
30 cm = side of the tile.

Ans: The side of a square tile with perimeter 120 cm is 30 cm.

HOTS Questions

1. We can have four different kinds of rectangles with area of 24 sq units. As we can get 24 units in four ways (recalling the factors of 24). Thus, One rectangle can have a single row of 24 squares, the second rectangle can have two rows of 12 squares, the third rectangle can have 3 rows with eight squares each and the fourth rectangle shall have four rows with six squares each.



2. Athletes don't start from the same position on the tracks because the perimeter of the tracks increases from inner to outermost side. In order to provide a level-playing field to all athletes the positions are marked in such a manner that all the contestants run the same distance.

Everyday Maths

1. The length of the wire required for fencing = Perimeter of the garden
= Sum of all sides = 8 m + 8 m + 4 m + 4 m + 6 m = 30 m

Ans: Razia needs 30 m of barbed wire for fencing the garden.

2. Cost of one metre wire = Rs 18
cost of 30 m of wire = Rs 18 x 30 = Rs 540

Ans: The cost of wire needed for fencing is Rs 540

3. In 2 metres number of trees planted is 1
in 30 m number of trees planted is: $\frac{1}{2} \times 30 = 15$

Ans: Razia would plant 15 trees.